

PART A - (MEDIX)

Medical Employee Management and Scheduling Application/Web Software (MEDIX)

Please type your name and label the deliverable # and section (e.g.: 2.1, 2.2,). Write all your work in this collective document with appropriate labels.

Four deliverables

Deliverable #1 (3 Marks) due week 2 for professor review and feedback Create an overall description for your problem domain/opportunity, fill in as appropriate the following:

Section 1 “Introduction” and sub sections:

- 1.1 Hospitals are critical healthcare institutions which face many unique challenges daily. With increasing patient rates, constant hiring, and the current large database of patients and medical staff, hospitals would benefit from an application which integrates specific utilities to keep a more organized and accessible format. Our project aims to outline the scope of the importance of developing an easily accessible and effective application for hospital staff.
- 1.2 **Purpose:** Identify the product whose software requirements are specified in this document, including the revision or release number. Describe the scope of the product that is covered by this SRS, particularly if this SRS describes only part of the system or a single subsystem.

Medix is an employee management and scheduling application software that does more than just save you time on scheduling. It also helps you improve communication, eliminate excuses, boost accountability among your staff, and track time and attendance. This software application will solve the problem of Talent Acquisition - Applicant tracking system, automated background screenings, electronic onboarding & hiring analytics • Workforce Management - Time & attendance, employee scheduling labor management, overtime monitoring & communication system • Employee Engagement & Retention - Staff surveys, performance monitoring and so much more. This is version 1.0.1. An improvement on version 1.0.0.

1.2 Document Conventions: Describe any standards or typographical conventions that were followed when writing this SRS, such as fonts or highlighting that have special

significance. For example, state whether priorities for higher-level requirements are assumed to be inherited by detailed requirement.

[FATIMAH]1- Table of Acronyms throughout this document:

Acronyms	Description
API	Application Program Interface
IT	Information Technology
HR	Human Resources
FR	Functional Requirements
NFR	Non-Functional Requirements
SQL	Sequential Query Language

PHP	Hypertext Preprocessor
JSON	JavaScript Object Notation
JS	JavaScript
PIPEDA	Personal Information Protection and Electronic Documents Act
ID	Identification
AWS	Amazon Web Services
TFS	Transport Layer Security
MFA	Mobile Two-Step Verification
HTTP	Hyper Text Transfer Protocol Secure
GPS	Global Positioning System

2- The use case numbering inherits from the first digit of the functional requirements.

1.3 Intended Audience and Reading Suggestions:

Describe the several types of reader that the document is intended for, such as developers, project managers, marketing staff, **end users** (varies based on project and especially important to list), testers, and documentation writers.

Developers:

Mobile Application Software Developer: These a professional who specializes in creating software applications specifically designed for mobile devices. These developers are skilled in programming languages, frameworks, and tools that are relevant to mobile app development. Their primary focus is on building applications that run on smartphones, tablets, and other mobile devices, catering to various platforms such as iOS, Android, or cross-platform solutions.

Web application software developer: These are professionals who specialize in creating software applications that operate on web servers and are accessed through web browsers. These developers are skilled in web programming languages, frameworks, and tools to design, develop, and maintain dynamic and interactive web applications. Their focus is on creating applications that run on the internet and can be accessed by users from various devices with a web browser.

PROJECT MANAGERS: Project managers organize software projects and can assign tasks to software engineering teams according to the specifications of a task. Software project managers use their leadership skills, technical knowledge, and experience to direct their teams and ensure software meets client requirements.

PROJECT TESTERS: They are technically skilled members responsible for ensuring that all the software functions operate smoothly, no bugs are present in the final product and help identify any holes in the program's security. They work within the quality assurance function of a computer software or programming team, striving to ensure superior performance and reduce the occurrence of issues.

MARKETING STAFF: Software marketing staff are professionals responsible for promoting and selling software products or services. They play a crucial role in the success of a software product by developing and implementing marketing strategies to reach and attract target customers.

END USERS: End users are the clients the software application is being developed for. In this case, the end users are the hospital management board and its employees.

SAMUEL: 1.4 PROJECT SCOPE:

Hospitals require a large staff to keep the institution running successfully, with proper patient care, technological support, scheduling, all while following regulations, safety procedures, and privacy mandates.

From recruitment to retention, employee management and everything in between, Medix employee management application software provides a platform that transforms the

relationship between healthcare organizations and their employees. Our innovative approach to recruitment, hiring, workforce management, pay and engagement fosters a culture where people want to work. Healthcare organizations can use Medix software to dramatically reduce turnover rates, decrease costs and improve the quality and continuity of healthcare work. We are focused on redefining the employee experience, while offering managers actionable data and insights throughout the entire employee journey—all from one unified software platform. Purpose-built for healthcare organizations, Medix integrated suite of software features: • Talent Acquisition - Applicant tracking system, automated background screenings, electronic onboarding & hiring analytics • Workforce Management - Time & attendance, employee scheduling labor management, overtime monitoring & communication system • Employee Engagement & Retention - Staff surveys, performance monitoring, automated rewards system, new hire tracking, financial wellness & early wage access for employees • Reporting & Compliance - Overtime tracking, time theft monitoring, employee utilization, hiring analytics & payroll- based journal reporting.

IN SCOPE	OUT SCOPE
Shift Scheduling: Implementing a scheduling feature to manage employee shifts, including the ability to assign shifts, view schedules, and handle shift swaps.	Extensive Reporting Beyond HR Metrics: Limiting reporting features to HR-related metrics and excluding extensive reporting on medical or patient-related data.
Employee Information Management: Designing a module to store and manage essential employee information such as personal details, contact information, and work history.	Patient Information Management: Excluding features related to patient information management, as the focus is on employee management.
Leave Management: Developing a leave management system to allow employees to request and managers to approve or	Medical Records System: Not including functionalities for managing patient medical records, as it falls outside the scope of

deny leaves, with appropriate notifications.	employee management.
Training and Certification Tracking: Creating a module to track employees' training programs and certifications, ensuring compliance with industry standards	Pharmacy Management: Excluding functionalities related to pharmacy management or drug inventory control, as it is outside the scope of the project.
Performance Reviews: Designing a system to conduct regular performance reviews, including setting goals, feedback collection, and performance evaluations.	Medical Billing System: Not incorporating features related to medical billing or invoicing patients, as it's not within the scope of employee management.
Health and Safety Compliance: Ensuring the software supports tracking and compliance with health and safety regulations, including certifications and training related to safety procedures.	Clinical Decision Support: Excluding clinical decision support features for healthcare professionals, as the software is designed for employee management.
Employee Self-Service Portal: Providing a self-service portal for employees to access their information, submit requests, and view relevant documents.	Integration with Non-HR Systems: Avoiding integration with systems not related to HR functions, such as equipment management or facility maintenance systems.
Payroll Integration:	Patient Appointment Scheduling:

Integrating with the hospital's payroll system to automate salary calculations, tax deductions, and other payroll-related tasks	Excluding features for scheduling patient appointments, as the software is focused on managing hospital employees.
Employee Onboarding: Developing a process for new employee onboarding, including documentation, training modules, and orientation schedules.	Patient Billing and Invoicing: Excluding features related to patient billing and invoicing, as it is not part of employee management.
Performance Appraisals: Creating a system for conducting regular performance appraisals and providing feedback to employees.	Medical Diagnosis Tools: Not incorporating medical diagnosis tools or decision support systems for healthcare professionals.
Workforce Analytics: Implementing analytics tools to analyze workforce trends, employee turnover, and other relevant metrics.	Clinical Research Management: Excluding functionalities related to managing clinical research projects, as it goes beyond employee management.
Staff Attendance Tracking: Developing a feature to track and manage employee attendance, including clock-in/out functionalities.	Patient Appointment Scheduling: Omitting features for scheduling patient appointments, as the focus is on hospital employee management.
Employee Benefits Management:	Pharmacy Inventory Control:

Designing a module to manage employee benefits such as health insurance, leave policies, and retirement plans.	Excluding features for managing pharmacy inventory or drug procurement.
Skill and Competency Tracking: Creating a system to track employees' skills and competencies, facilitating efficient resource allocation.	Patient Medical Records: Not including functionalities for managing patient medical records or healthcare history.
Employee Recognition Programs: Implementing features to recognize and reward employees for outstanding performance or achievements.	Medical Equipment Maintenance: Excluding features related to the maintenance of medical equipment in the hospital.
Shift Differentials: Integrating a system to calculate and manage shift differentials based on specific working hours.	Facility Security Systems: Omitting integration with security systems that are not related to employee management.
Succession Planning: Developing tools to identify and prepare employees for future leadership roles through succession planning.	Non-HR Financial Transactions: Avoiding features related to financial transactions unrelated to HR functions.
Staff Communication Channels:	External Vendor Management:

Implementing communication channels within the software for staff announcements, updates, and collaboration. Staff Training and Development: Managing training programs and development opportunities to enhance employee skills.	Excluding tools for managing relationships with external vendors not involved in HR processes. Hospital Inventory Control: Omitting features for managing overall hospital inventory and supplies.
Time-Off Accruals: Designing a system to calculate and manage time-off accruals for vacation, sick leave, and other types of leaves.	Patient Admission/Discharge: Not incorporating features for managing patient admission and discharge processes.
Employee Health and Wellness Programs: Integrating features to promote employee health and wellness, such as fitness challenges and wellness initiatives.	Hospital Billing System: Excluding functionalities related to the hospital's overall billing and revenue cycle.
Employee Surveys: Implementing surveys to gather feedback from employees regarding workplace satisfaction and engagement.	External Training Providers: External Entities Providing Training
Employee Grievance Handling:	Legal Document Management: Legal Documents Unrelated to Medical Employment

Creating a mechanism for employees to raise grievances, with a structured process for resolution.	
Task and Project Management: Including features for task assignment and project management within teams or departments.	Task and Project Management for Non-Work-Related Activities: Managing tasks and projects unrelated to work responsibilities or projects related to medical employee duties.
Employee Directory: Providing a comprehensive employee directory with contact information and organizational hierarchy.	Non-Medical Employee Directory: Creating a directory for individuals not directly associated with the medical employee workforce.
Training Resource Library: Developing a repository of training resources and materials accessible to employees.	Training Resource Library for Non-Work-Related Content: Including resources or materials that are not related to medical training or professional development.
Overtime Management: Integrating tools to monitor and manage employee overtime, ensuring compliance with labor regulations.	Overtime Management for Non-Medical Activities: Monitoring overtime for tasks or projects unrelated to medical duties.
Employee Exit Process:	Exit Process for Non-Employee Exits: Managing exit processes for individuals who are not medical employees within the

Creating a process for managing employee exits, including exit interviews and clearance procedures.	organization.
Overtime tracking: Overtime tracking involves monitoring and recording the hours that employees work beyond their regular scheduled working hours. This includes any extra time worked beyond the standard workday or workweek.	Specialized Healthcare Modules: Modules specific to medical specialties (e.g., radiology, pathology) are often handled by specialized software.
Time theft monitoring Time theft monitoring involves keeping an eye on employees who try to take advantage by not working when they should be. It helps ensure that people are doing their job during their designated work hours	Non-Employee HR Functions: Highly specialized or custom healthcare processes that are not common to general employee management

SAMUEL: SOME BASIC FEATURES OF MEDIX:

MANAGEMENT POINT OF VIEW:

- Team management - Onboard new employees, approve time off requests, review shift trades, and fill shifts in seconds.
- Team accountability - Shift confirmation, mobile clock in, shift task lists, team task lists, and chat ensure everyone is on the same page.
- Team communication - Notify your team of changes instantly and communicate with coworkers 1:1 or in group chats.

- Team empowerment - Your team can manage availability, submit time-off requests, and collaborate on shift trades.
- One-click scheduling - Build your entire work schedule in one click with Auto Scheduling.
- Confident shift coverage - Receive shift confirmations from employees and be notified immediately of swaps and drops.
- Share schedules - Publish the schedule and your whole team is instantly notified of their shifts.
- Labor forecasting - Easy-to-use labor tools help you manage your labor budget efficiently.
- Control overtime costs - Get overtime alerts and labor reporting to make managing labor costs easier. • Schedule enforcement - Limit clock ins to shifts and locations to ensure that everyone is where they need to be at the right time.
- Payroll integration - Import your timesheets to process payroll quickly and accurately.
- Labor reports - Use labor distribution to manage costs and forecast for the future.

- Easy to use time clock to clock in and out
- Message with staff 1:1 or in groups
- Monitor and approve shift trades and time off requests
- View the entire staff schedule anywhere at anytime
- Create a work calendar to efficiently plan schedules
- Quickly alert staff of changes and updates to the schedule
- Messaging and scheduling permissions for easier management
- Geofencing features to prevent buddy punching

EMPLOYEES POINT OF VIEW:

- Clock in and out of shifts from the app

- Instantly trade shifts with coworkers
- View work schedules from anywhere at anytime
- Pick up extra hours with open shifts
- Check the work calendar to easily request time off
- Privately message coworkers 1:1 or in groups within the app
- Easily check who's working which shifts

1.5 References: List any references/websites you used to research the project problem.

1. Google playstore(2024.)
https://play.google.com/store/apps/details?id=com.thisclicks.wiw&hl=en_CA&gl=US&pli=1
2. G2.com.Inc (2024) <https://www.g2.com/categories/medical-staff-scheduling>
3. Apple Inc. (2024) <https://apps.apple.com/us/app/onshift-mobile/id600755207>
- 4 Amazon Web Services, Inc (2023) https://aws.amazon.com/security/?nc1=f_cc
5. Indeed (2024) <https://www.indeed.com/career-advice/finding-a-job/project-manager-in-software#:~:text=Project%20managers%20in%20software%20organize,ensure%20software%20meets%20client%20requirements.>
6. BetterHealth (2024)
<https://www.betterhealth.vic.gov.au/health/servicesandsupport/hospital-staff-roles>

Fazlur: Section 2 “Overall description” and sub-sections:

This software is a standalone product that was developed specifically to meet the unique challenges and prospects in the medical services. It creates an independent product that is intended to change patient care and simplify all processes associated with medicine. However, it is a unique and innovative solution that specifically addresses the emerging requirements of healthcare providers.

2.1 Product Perspective

- This is a new software.

- The system will be developed in a user-friendly way, which is developed to maintain the data of the hospital management system and organize the system from the older one.
- The software will coordinate with all other information of the medical when the administration needs to make any decision.
- The system will provide and introduce some unique features which do not exist in the running system.
- A user-friendly interface will be there which will attract the stakeholders, the patients, and the staff. Moreover, the interface will attract all who are the user of this system.
- The system will align with the industry regulations and ensure the secure handling of patient data.
- Maintenance and support will be very user-friendly. So whenever the new developer is assigned, he/she can upgrade the system as per the demands.
- The security and safety of the system will be updated from the previous system. So that the system can secure the data of the user.

2.2 Key features of the software:

The major functions of the software are summarized below:

- Electronic Health Record system
- Patient appointment (New/Old) system
- Medication and follow-up reminders for all patients
- Providing a secure communication channel between patients and health care providers.
- Connectivity with the devices.
- ID cards record storage and create new IDs.
- Connectivity with the Google Maps
- Payment channel for patients and parking.
- Patient admission form
- Employees personal form
- Electronic discharge certificate
- Apps login using face app
- Hospital notice board
- Hospital Campaign of summer medical camp
- Hospital database from the beginning (by upgrading as the hospital has)

2.3 User Classes and Characteristics

Identified user classes include:

- Patients: Utilize the application for personal health management.

- Healthcare providers: Access patient data, schedule appointments, and communicate with colleagues.
- Administrators: Manage system settings and user access.
- All the staff Who have permission to access.
- Insurance Providers give claims transmission in addition to coverage verification and billing for patient information.
- Emergency Responders Speedily have vital data regarding patients' record in records of emergencies.
- Remote Monitoring Teams monitor and analyze real-time patient data for intelligent remote patient or condition management.
- Hospital IT Support should be managed are technical issues, system updates, and user support concerning the application.

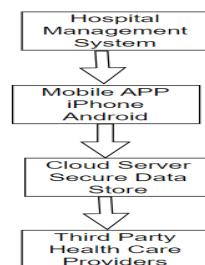
2.4 Operating Environment

- The system will operate from any reliable internet connection and browser:
- The software will be available and usable for both iOS(iPhone) and Android Platforms.
- Background data storage will be facilitated by a secure server.
- To get the best performance on all platforms, the application will make use of native development technologies.
- Offline Mode: Implement an offline mode, allowing users to access critical functionalities even when there's no internet connection. The system can sync data once a reliable connection is restored.
- Device-Responsive Design: Emphasize a device-responsive design to ensure that the software is accessible and user-friendly on various screen sizes, including tablets and different smartphones.
- As a furtherance of account security, multi-factor authentication will be implemented.
- The system will be regularly audited and updated security and upgraded each time when addressing the vulnerabilities and ensuring the system's security.
- Users will also be able to get comprehensive user support service through online documentation, FAQs, and a help desk so they can resolve their issue and better understand how the software is supposed to work.
- Daily backups of user data would be carried out to ensure that when the system fails or when unexpected events arise, which could cause data loss, the data could be restored.
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2.5 Assumptions and Dependencies

- Assumptions: If the user has only reliable internet connectivity for real-time data synchronization.

- Dependency: Integration with third-party API s data retrieval.
- Regular Software Updates: Assume that users will regularly update their browsers and mobile operating systems to the latest versions, ensuring compatibility with the evolving features and security measures of the hospital management software.
- Consistent Mobile App Updates: Assume that users of the mobile applications on iOS and Android platforms will regularly update the apps to benefit from new features, improvements, and security patches



YUSUF NASIMI

Section 3 External Interface Requirements

3.1 User Interface

Users will be able to access our system by using our mobile app. This app will be supported on Apple and Android, with a web server giving content to both web browsers and mobile applications. Some supported browsers would be Chrome, Safari, Edge, Opera and Firefox. By using this approach, end users will be able to access the application from multiple, easily accessible platforms with internet connection. The interface can potentially use JS, CSS, and HTML for the front-end, and PHP, SQL, and JSON for back-end (Fatimah).

Commented [YN1]: I need to add more to the 3.2 and 3.3 list and make a system context diagram for 3.1

Commented [YN2R1]: Might need to change some peoples stakeholder positions as well

Commented [FR3R1]: I added some stuff in blue let me know what you think

Commented [YN4R1]: @Fatimah Rana Alright I like it I just thought we were making a mobile app

Commented [FR5R1]: Mobile apps are also able to be accessed on the web if a web server is created. Makes it more versatile and user-friendly.

Commented [YN6R1]: I have added a diagram

Commented [FR7R1]: I have edited the diagram

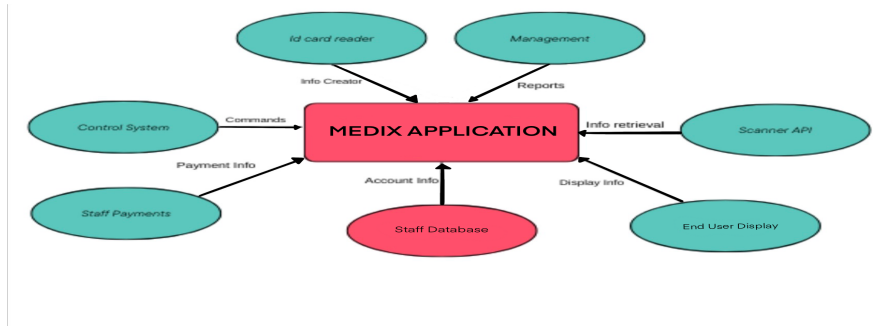
Commented [YN8R1]: Looks good

Commented [YN9R1]: how much do we need for 3.2 and 3.3

Commented [FR10R1]: i think for 3.2 its good but i think we might be missing some for 3.3

Commented [YN11R1]: I think we're good on 3.3, we only need to list third party software

Commented [YN12R1]: also the stuff i added to 3.3 are highlighted blue



3.2 Hardware Interface

ID card reader

Scanner API

Biometric

Printer

Cameras

Speakers

Ambient light sensors

High resolution displays

Fingerprint Scanner

Face recognition

Calendar

Work Schedule

Schedule changes

Direct and group messaging

Notifications

Staff Index/Sorting

User Interface

Windows

Mac

Linux

Payroll Integration:

Interac

PayPal

VISA

Mastercard

Deliverable #2

Stakeholder Register					
Stakeholder Name	Stakeholder Position	External/Internal	Stakeholder contact details	Operational/Executive	Interest (high, medium, low)
Samuel Osoata	Marketing officer	Internal	osoata1@my.cennialcollege.ca	Operational	medium
Yusuf Nasimi	IT administrator	Internal	ynasimi@my.cennialcollege.ca	Operational	high
Mary Jane	Customer	External	MaryJane@gmail.com	Operational	high
Bruce Musk	Investor	External	BruceMusk@gmail.com	Executive	high
Fazlur Rahman	IT security Officer	Internal	frahma48@my.cennialcollege.ca	Operational	low

Fatimah Rana	Programmer	Internal	frana2@my.centennialcollege.ca	Operational	Medium
Joe Nimble	Director	Internal	JoeNimble@outlook.com	Executive	high
William John	Project manager	Internal	Williamjohn@gmail.com	Executive	high
Chris Owen	Hospital Administrator	Internal	ChrisOwen@gamil.com	Executive	high
Dan Brown	Operations Manager	Internal	DanBrown@gmail.com	Executive	high
Taylor Barbara	IT Management	Internal	TaylorBarbara@gmail.com	Operational	high
Bill Bezos	HR Manager	Internal	BillBezoz@gmail.com	Executive	high
Steven Tam	Legal Team	Internal	stam@inq.law	Executive	High
Phill Basil	Database Manager	Internal	PhillBasil@gmail.com	Operational	high
Dr. Andrew Park	Ontario Medical Association	Internal	Info@oma.org	Executive	High
Dr. Michelle Webb	Doctor Representative (End User)	Internal	Michellewebb@gmail.com	Executive	low
Roger Daniels	Nurse (End User)	Internal	RDaniels@gmail.com	Executive	low
Paige Stanley	Practitioner (End User)	Internal	Paigestanley@gmail.com	Executive	low

Hospital Administrators

PIPEDA

Canadian Health Admin

Etc.

Deliverable #3 (2 mark) due week 3 for professor review and feedback

Prepare interview questions that you as a software engineer will ask the stakeholders. The questions should clarify issues surrounding the applications that you are developing. To carry out this exercise, carry out a role-play among the team members where one of the team members would assume the role of a stakeholder and the rest of the team would assume the roles of the software engineering teams. Repeat the role play at least four times to gain good insight and record all questions and answers on the question template.

A minimum of 5 questions per round should be recorded. Add the output to Appendix D of your SRS document.

QUESTION	STAKEHOLDER	ANSWER
How is private/sensitive patient information handled in the system?	Hospital Administrator	Encryption and staff roles will facilitate who can access certain information.
What roles and permissions are included?	Hospital Administrator	Roles include doctors who can exclusively access patient records, nurses who can update patient charts, and administrative staff who regulate scheduling and billing.
Will other hospital systems, such as payroll, HR systems, and timekeeping be integrated into the new application?	HR Manager	Yes, some previous systems will be integrated into the new system, including timekeeping and pay rates to create an ideal Unified System. Payroll will stay separate.
How is the system currently managing healthcare professionals' shift scheduling and coordination?	HR Manager	Currently, the scheduling is done manually. As the app progresses, after integrating the previous manual scheduling system, we can overcome challenges in ensuring proper staffing numbers during peak hours. A more automated approach would be ideal.
In the context of reporting and analytics, what kinds of insights are you hoping to draw from the new system	HR Manager	Reports on resource usage, patient satisfaction, and staff performance are required. This will support

to help with resource planning and decision-making?		our efforts to optimize operations.
What specific compliance requirements is the system adhering to?	HR Manager	Our system will follow the Personal Information Protection and Electronic Documents Act (PIPEDA) and other relevant healthcare data protection standards
To guarantee that only those with permission can access sensitive data, how is the existing system managing user authentication and access control?	IT Management	Using a role-based system with personalized access control systems makes it so only those authorized can access data. Multi-factor authentications will help with stricter access monitoring
How would the system handle forgotten password or credentials?	IT Management	By having an IT team to help manually reset passwords, we can complete the task. However, for more efficient work, there are self-serve password change options, as well as ID cards and scanners which keep your credentials.
How will we use the money?	Investor	We will use the money by keeping our database running and making constant improvements to our database.
What are the biggest risks?	Investor	Our biggest risks are hospitals not wanting to make a deal with us to implement our database into the hospital.
What is the speed of the database	Database Manager	Our databases speed is more than enough to upload information in a matter of seconds
What is the storage size of the database	Database Manager	Our databases size is 5tb

What issues does the parking enforcement team currently have with managing parking rules, and where do you think improvements or innovations could be made in the system or process?	Parking Enforcement	Parking enforcement faces delays with paper-based processes. Real-time data integration is needed for efficient enforcement. Mobile capabilities and data analytics can enhance system effectiveness.
Can you elaborate on the expected frequency of updates for the software and how the system will communicate these updates to users to ensure they stay current?	Hospital administrator	A quarterly update schedule, with minor improvements and bug fixes released more frequently, as needed.
Regarding integration with wearable health devices, what specific types of devices are you targeting, and what kind of data should the system retrieve from these devices?	Hospital Administrator	Our primary focus is on integrating with widely used wearable health devices such as fitness trackers, smartwatches, and medical monitoring devices. We aim to retrieve essential health data from these devices, including heart rate, activity levels, sleep patterns, and, if available, more advanced metrics like blood pressure and oxygen saturation.

Deliverable #4 (4 mark) due week 4 for professor review and feedback

Based on output of deliverables #1 and #2 create two lists of requirements. The first list should reflect functional requirements (FR) and the second list should reflect non-functional (NRF) requirements. Per requirement identify the following:

1. provide an ID# for functional FR01, FR02...etc. for non-functional NFR01, NFR02, ..etc.
2. A requirement title.
3. Write a small paragraph to describe each requirement.
4. A priority per requirement use quality deployment functionality (QDF) technique to assign priorities

5. A requirement originator (i.e. stakeholder who requested it)

Functional Requirements list				
Requirement ID	Requirement title	Short Description	Priority	Requester
FR 001	Scheduling	The application must allow hospitals to create, edit, delete, and organize staff schedules either manually, or with auto-generated features. Staff should also be able to view their schedules at any time.	Required	Hospital Administrator
FR 002	Staff Registration	The application must allow access for staff/admin to register and log in/out of the system regularly.	Required	Hospital Administrator
FR 003	Staff ID Name	The application will have staff members' names visible and accurate on the account.	Required	Hospital Administrator

FR 004	Staff ID Number	The application will have all staff members' ID # visible and accurate on the account.	Required	Hospital Administrator
FR 005	Staff ID Photo	The application will have all staff members' facial identification photos visible and accurate on the account.	Required	Hospital Administrator
FR 006	Staff Roles	The application must display the role of the staff member and be visible and accurate on the account.	Required	Hospital Administrator
FR 007	Medical Staff Role	The application must address the roles of all doctors, nurses, and allied health professionals within the corporation and must be sorted and assigned as	Required	Hospital Administrators /IT services

		“Medical Staff,” giving limited web access depending on the specific employee ID, and no admin access to the app.		
FR 008	Hospital Support Staff Role	The application will store roles of all clinical assistants, patient services, volunteers, front desk workers, and porters must be sorted and assigned as “Hospital Support,” giving limited web-access depending on the specific employee ID, and no admin access to the app.	Required	Hospital Administrators
FR 009	Scanner API	The application must include a program/API to activate ID cards and scanners, which hold	Required	Hospital Administrators

		each individual staff member's account, status, permissions, and allow entry to rooms.		
FR 010	Push/Banner notifications	The application should have notifications shown to the user for any updates, events, maintenance time, scheduling, etc.	Medium	End Users
FR 011	Staff Communication	The application might allow text-based communication between all staff members and will be useful for quick communication access across the institution. Management can also post larger announcements.	Medium	Hospital Administration /End users
FR 012	Staff Requests	The application	High	End Users

		must allow staff of the same title to request switch shifts and send their schedule availability to the bookkeepers.		
FR 013	Schedule Views	This system will allow you to view and keep track of the staff's schedule.	Medium	End Users
FR 014	Task View	The application could include a 'task view' screen with multiple layouts for end users to see their pending tasks.	Medium	End Users
FR 015	GPS based Clock-in technology	The application will include a Clock-in system, allowing employees to clock in remotely ONLY when detected at the facility with GPS, at any scheduled time.	High	Hospital Administrator

FR 016	Paystub/spreadsheet distribution	The application should allow end-users to see their paystub virtually or through a spreadsheet.	Medium	Finance Department
FR 017	Attendance Report	The application should provide attendance reports for administrators to track/analyze the staff attendance. Insights into attendance data can support processes within schedule building.	High	Hospital Administrator
FR 018	Time-Off Management	The application must facilitate effective time-off or personal leave management, allowing staff to submit time-off requests. Administrators must have the ability to accept,	High	HR

		decline, or inquire about the request.		
FR 019	Training and Certification Tracking	The application should provide a system for tracking staff and their certifications. These must be uploaded by the staff member and verified by administrators and able to input and update the certifications.	Medium	HR
FR 020	Emergency Contact Information	The application should have a secure and accessible repository of staff member's emergency contact, which is self-input by the staff member. This ensures safety for crucial incidents.	Medium	End Users
FR 021	Personalized Dashboards	The application will be designed so staff members	Medium	End User

		can have personalized dashboards displaying relevant information such as pending requests, scheduled appointments, and upcoming shifts.		
FR 022	Telemedicine Integration	The application might help if staff needs to consult with doctor after discharge	Medium	End User
FR 023	Payment Channel	The application will show any charges from staff and allow payment through the application.	High	Finance Department
FR 024	Parking Management	The application will maintain traffic and security.	High	Parking Enforcement
FR 025	Connectivity to 999	The application will provide options if any medical emergency anyone from	High	End Users

		the office staff can easily connect to 999		
FR 026	Diagnostic Imaging Integration	The application must integrate seamlessly with diagnostic imaging systems and allow medical staff to access and view diagnostic images.	High	Hospital Administration
FR 027	Laboratory Test Result	The application will automatically take a report and send to the consent doctor and the department after releasing.	High	
FR 029	Multi-Language support	The application must allow staff from various regions to use the app and adhere to the language.	High	End users
FR 029	Insurance verification and Claims Processing	The application might allow authorized personnel to see if any staff have insurance	medium	End-Users

FR 030	Staff Directory	The application should include a navigable staff directory with contact details, role, department info	Medium	End Users
FR 031	Internal messaging system	The application will include an internal messaging system for secure communication between staff	Medium	End Users
FR 032	Performance Evaluation	The application could provide forms/surveys about performance, providing valuable feedback for continuous improvement.	High	HR
FR 033	Staff Sorting Matrix	The application should provide a flexible staff sorting matrix/index, allowing the user to filter, categorize, and organize staff members	Medium	Hospital Administration

		based on certain criteria such as name, qualifications, title.		
FR 034	On-Call Scheduling	The application must support the creation of on-call schedules for medical staff, allowing the institution to seamlessly manage emergency situations.	High	Hospital Administrators
FR 035	Incident Reporting	The application should include an incident report feature, allowing staff to report issues within the hospital and receive quicker responses.	Medium	Incident Management Team
FR 036	Internal Job Posting	The application could facilitate job postings, allowing staff to visit external links and post job listings for co-workers to span career	Medium	Human Resources

		options within the industry.		
FR 037	Employee Health Records	The application should maintain up-to-date records of the staff's medical health records such as screenings and relevant health information.	Medium	HR
FR 038	Remote Work Management	The application should support remote work management for those in the facility working remotely, allowing for coordination and to continue operations from a distance.	Medium	HR

Non-Functional Requirements list				
Requirement ID	Requirement title	Short Description	Priority	Requester

NFR01	Security/protocols	This application must encrypt all confidential information and only accessible to specified admin. Additionally, healthcare privacy regulations and rules must be followed.	Mandatory	Healthcare Administration + IT Security
NFR02	Usability	The application must ensure a user-friendly interface for easy access across the features and minimal confusion/errors.	High	End User (hospital staff/admin)
NFR03	Bandwidth	The application should uphold speed and high performance and maintain the performance with many active end users.	High	End Users
NFR04	Compatibility	This application should be able to meet minimum hardware	high	End Users

		requirements and be supported on most operating systems.		
NFR005	Reliability	The application must have a very low chance of failure rate such as 0.0002%	High	End users
NFR006	User Authentication	The application must be able to verify users with keycards and facial recognition	High	Hospital Administration
NFR007	Accessibility	The application should be accessible to people with the right clearance to comprehend the database in all forms (text-to-speech, color-blind friendly)	High	End Users
NFR008	Calendar Integration	The application must integrate a calendar to allow staff members to view and manage	Medium	End Users

		events/appointments/meetings		
NFR009	Data Recovery/Backup	The application must have a feature to recover data and to backup said data.	High	Hospital Administrators
NFR010	System documentation	The application should have documents showing how the application and its parts work.	Medium	IT Department/Hospital Administrators
NFR011	Regulatory Compliance	The application must comply with health regulations and other privacy laws.	High	Legal Team
NFR012	Responsiveness	The application must be very responsive with 5< seconds to be responsive	High	End Users
NFR013	Disaster Recovery Plan	The application must have a plan to recover from power outages, natural	High	Hospital Administrators

		disasters, and cyber-attacks.		
NFR 014	Localization	This application should support the English language and all variations.	Medium	End Users
NFR0 15	System Updates	The application must have a way to get future system updates through the internet	High	IT department
NFR0 16	Network Security	The application must have extremely high network security to protect staff members	High	End Users
NFR0 17	Usability Tests	The application must have ways to test if it is working correctly.	High	IT department
NFR0 18	Data Encryption	The application must have a way to encrypt data for further network security	High	IT Department
NFR 019	Confidentiality	The application	High	HR

		should only be accessible from authorized people		
NFR 020	Cross-Browser Compatibility	The application must support the use of multiple browsers such as Firefox and chrome.	High	End Users
NFR 021	System Backup Frequency	The application must make a backup of all data frequently	High	IT Deoartment
NFR 022	Data Integrity	The application must have a way to verify the integrity of the data.	High	HR/Hospital Administrators
NFR 023	Capacity	The application must have enough capacity to hold information on all our staff and future staff.	High	Hospital Administration
NFR 024	Mobile Performance	The application must have low requirements to run on phones to	High	End Users

		support high performance		
NFR 025	Manageability	The application must be easy to manage for people with access to the database.	High	End Users
NFR 026	Application Handbook	The application should include an information/handbook section for inexperienced users to gain familiarity with the app	High	End-Users/IT Department
NFR 027	Incident Response Time	The application must have a target time to address and resolve technical or security incidents to quickly remove threats and bugs.	High	End-Users/IT Department
NFR 028	Emergency Access	A designated emergency access feature should be in place, allowing authorized personnel quick access to critical	High	Hospital Administration

		information during emergencies.		
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